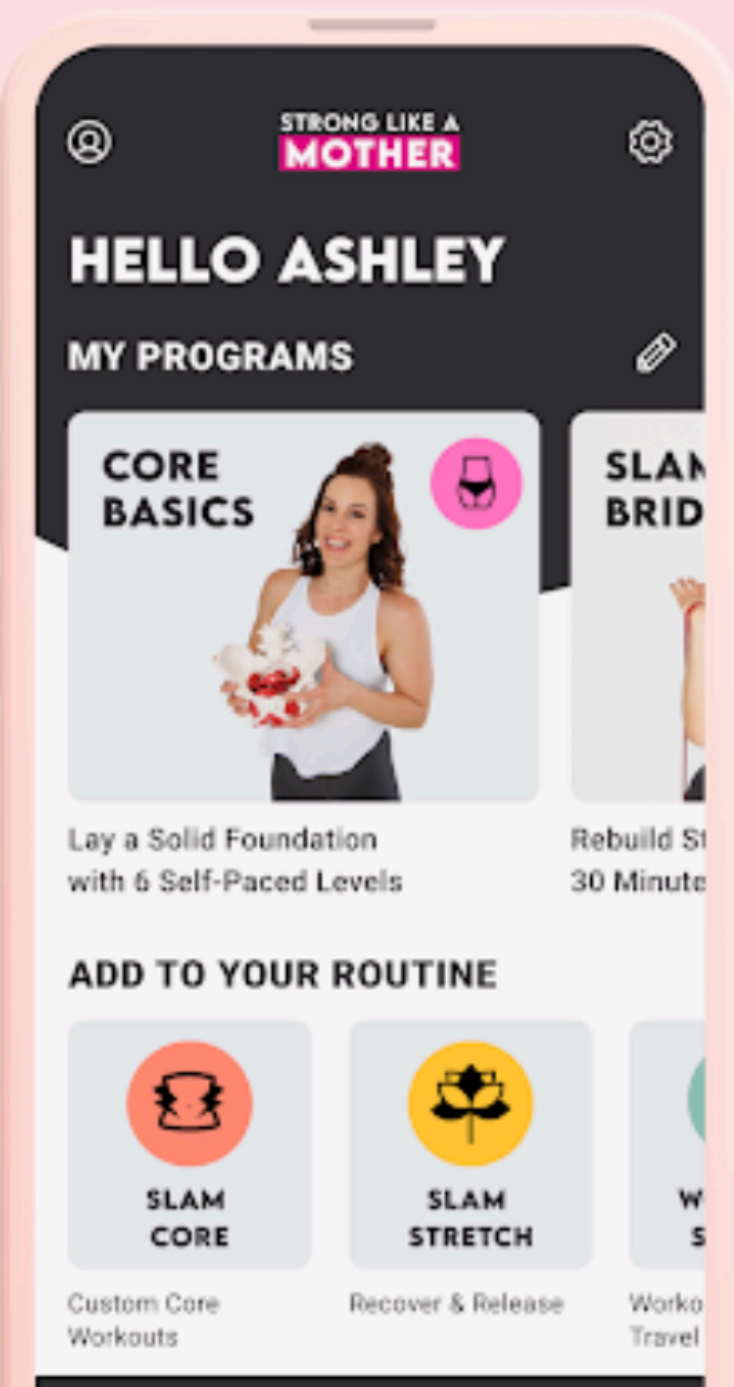


Matru-Kamal SWASTH HACKATHON 2026 • APP DESIGN

SWASTH (Maternal Health)

The Daily Cultural Companion for Mothers & Caregivers: 40-Week Growth, 4Q Garbh Sanskar Routines & Long-Term Care Continuity

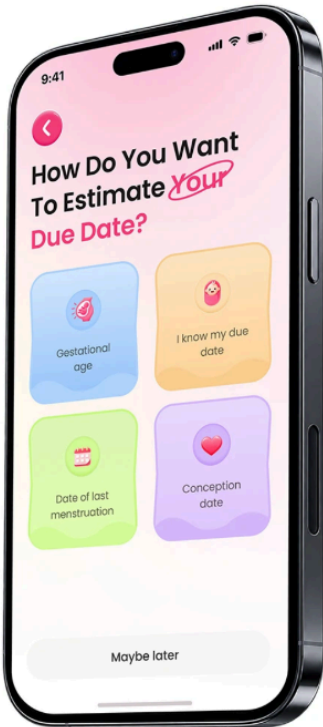
- 🌸 **Maternal & Women's Health Track** · 1,000-Day Care Continuity
- 📱 **Modern Dual Architecture:** Flutter Mobile (60 FPS) + Responsive Web Simulation
- 🌺 **4Q Daily Habit Hook:** Spiritual, Physical, Intellectual & Emotional micro-rituals
- 🛡️ **Zero Clinical Diagnosis:** Strictly assistive with NVIDIA NeMo Guardrails
- 📶 **On-Device Privacy:** 100% offline-ready encrypted local SQLite storage



40-Week Interactive Timeline & Personalized Intent Setup

Due Date Estimation

Easily estimate your baby's due date using different methods and get a more personalized pregnancy timeline based on your information.



Baby Growth Tracking

Track your baby's growth week by week with helpful insights, milestones, and personalized updates for every stage of pregnancy.

Track Your Baby's Growth

Stay up to date thanks to weekly articles and baby size data



1. Multi-Method Due Date Setup

Supports Gestational age, LMP date, Conception date, or manual due date for instant personalization.

2. Tactile 40-Week Timeline

Interactive week scrubber with culturally relatable fruit analogies (Peach at Week 14, 8.7cm, 43g).

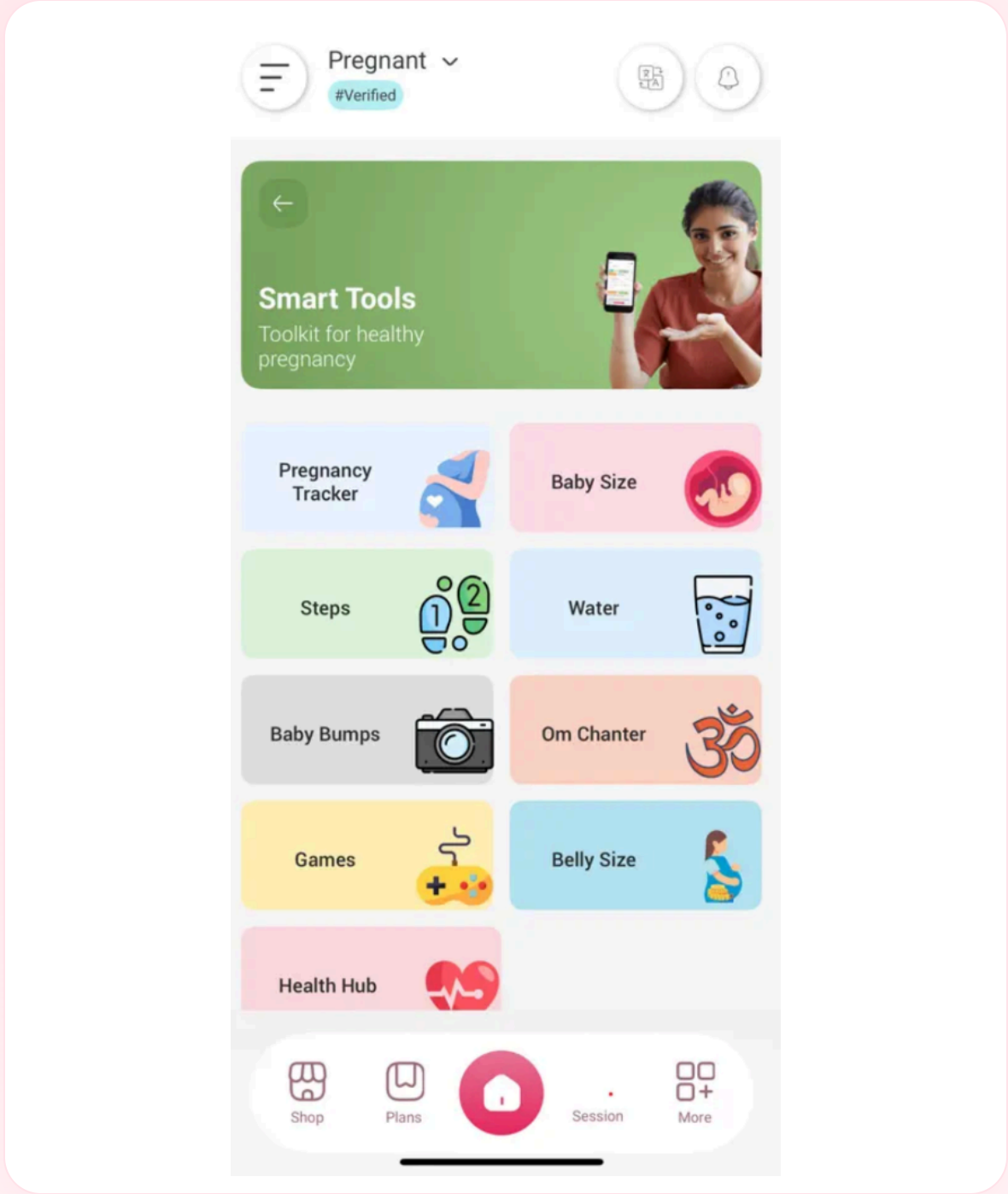
3. Gestational Countdown

Real-time Day 98 of 280 counter and upcoming primary health center checkup alerts.

4. Maternal Story Rings

One-tap vector SVG action rings for Hydration, Om Chanting, Birth Clubs, and Diet.

Interactive Health Toolkit & 4Q Garbh Sanskar Habit Engine



 Daily 4Q Activities

8-Tool Health Toolkit

Hydration 2,500ml tracker, Steps pedometer, 432Hz Om chanter, Kick counter, Health Hub BP/weight, Belly size, and TT vaccine tracker.

4Q Cultural Science Routine

12 daily habits categorized into Spiritual (SQ), Physical (PQ), Intellectual (IQ), and Emotional (EQ) quotients.

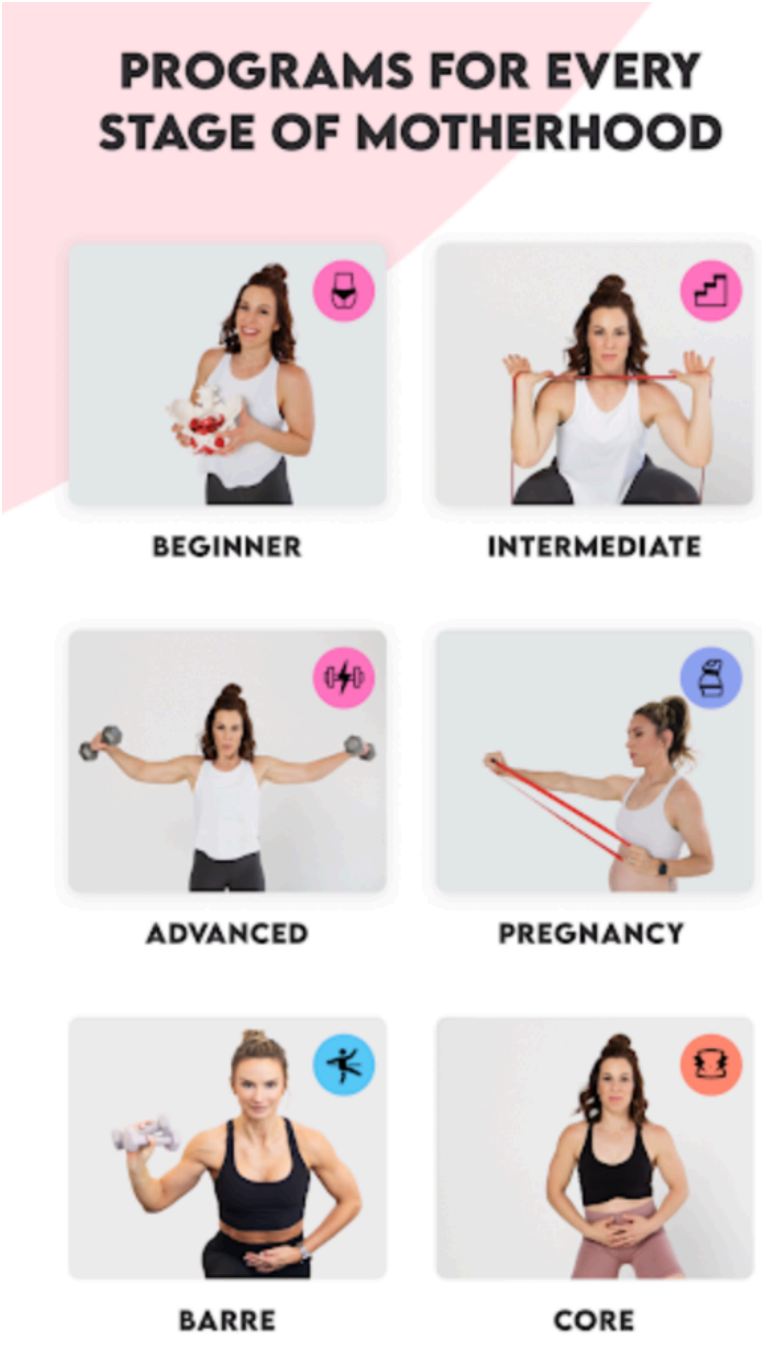
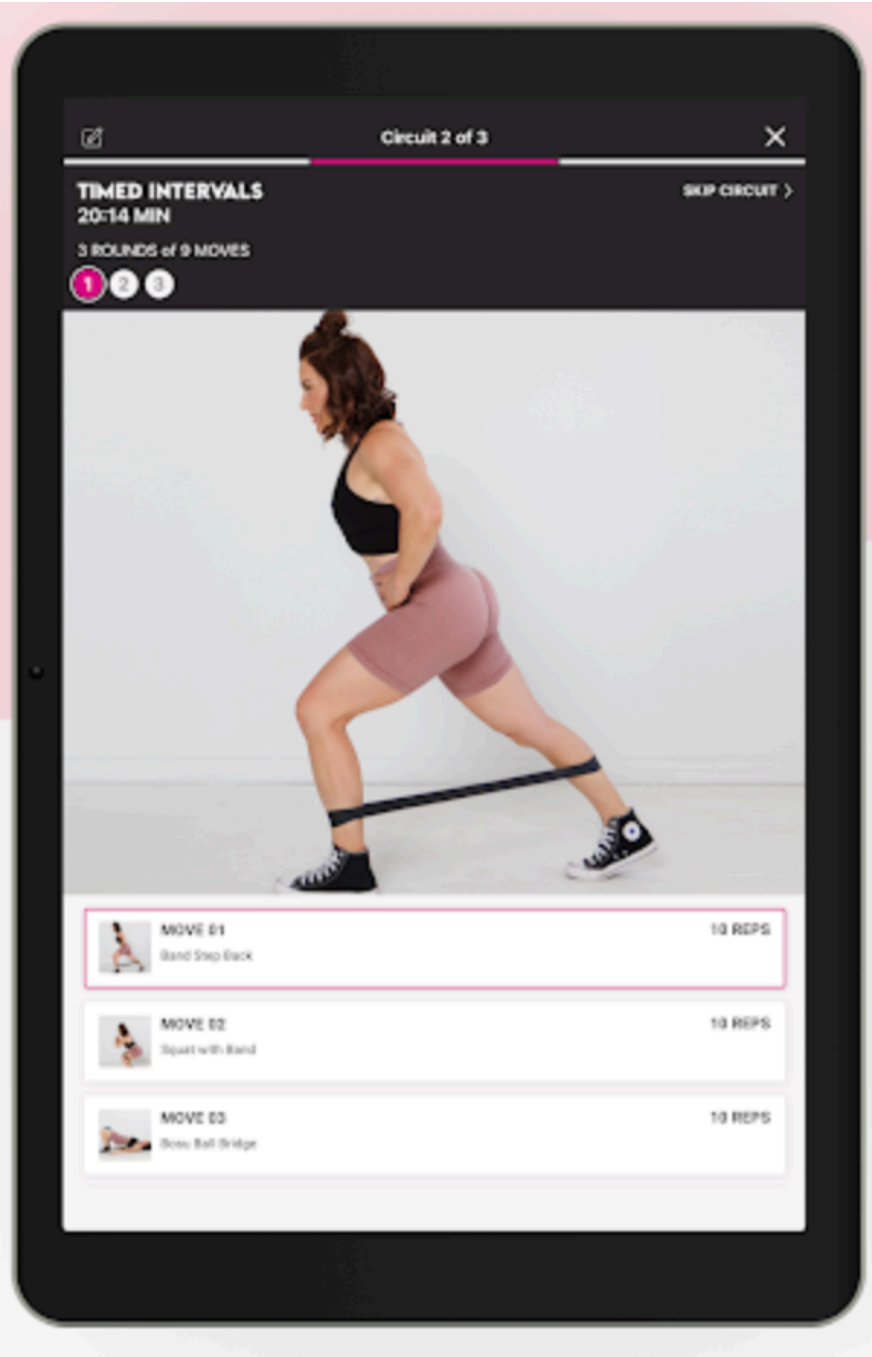
Real-Time Completion Sync

Dynamic progress card (8 of 12 completed, 67%) updating on-device SQLite database instantly.

Zero Emoji Standard

100% clean outline/duotone vector SVGs with pastel badge geometry.

Maternal Recovery Programs, Timed Circuits & 150+ Recipes



Programs for Every Stage

Beginner, Intermediate, Advanced, Pregnancy, Barre, and Diastasis-safe Core Rebuild programs.

Timed Interval Circuit Console

20:14 min guided workout player with 3 rounds of 9 moves, live countdown timer, and posture cues.

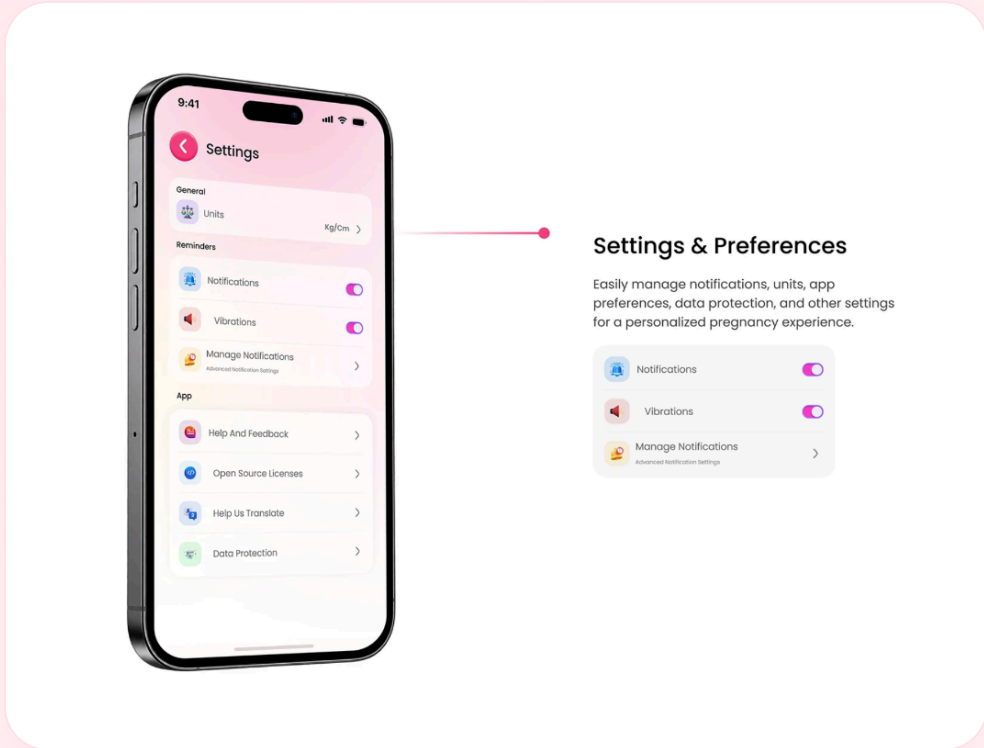
Specialist Mobility & Recovery

Pelvic floor release, prenatal yoga, and myofascial tension relief led by certified physios.

150+ Cultural Maternal Recipes

High-folate Moong salads, Kashmiri Saffron Golden Milk, and infant weaning Ragi mashes paired with Swasth AI.

Enterprise Architecture, ABHA Data Protection & 60-90 Day Impact



Open Enterprise Architecture

- **Flutter Mobile (iOS/Android):** 60 FPS fluid rendering with local SQLite persistence.
- **React/TypeScript Web:** Responsive standalone simulation portal.
- **Python (FastAPI):** Single unified backend with sub-5ms latency.
- **LangGraph + vLLM:** NVIDIA Nemotron and Llama 3 state-aware agent pipeline.
- **Indic NLP:** Whisper open voice notes & IndicTrans2 (10+ languages).
- **ABDM/ABHA Compliance:** Immutable consent-verified audit ledger.

Safety & 60-90 Day Impact

- **0% Diagnostic Advice:** Strict non-clinical scope with PHC referral routing.
- **NVIDIA NeMo Guardrails:** Intercepts severe symptoms and red flags instantly.
- **85% Drop in Administrative Burden:** Automated due lists for frontline ASHA workers.
- **90%+ Child Vaccine Adherence:** Drastically cuts the 6/10/14-week drop-off cliff.
- **Hyperlocal Birth Clubs:** Mutual peer guidance across 9,388+ online mothers.
- **100% Offline Functional:** Never blocked by low connectivity in rural centers.